

OSTASKOV, Russian Federation

WMO: 263890

Lat: 57.133N

Lon: 33.117E

Elev: 715

StdP: 14.32

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.5	-7.5	-17.6	2.2	-13.0	-11.7	3.0	-6.9	14.7	26.0	12.2	23.1	1.7	320	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	83.0	67.2	79.4	65.9	76.1	64.2	69.2	79.6	67.4	76.4	65.7	73.9	4.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	97.2	73.4	64.1	91.8	72.0	62.5	86.9	70.0	33.7	80.0	32.3	76.3	30.9	73.9	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 12.5	11.6	10.5		-20.6	87.1	5.6	4.5	-24.7	90.3	-27.9	93.0	-31.1	95.5	-35.1	98.8	
(5)			DB	-22.0	72.0	5.5	2.4	-25.9	73.7	-29.1	75.1	-32.1	76.5	-36.1	78.2	(p)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.3	18.9	19.2	27.5	41.8	52.5	60.2	65.7	60.9	51.7	41.1	30.7	23.7	
	DBStd	18.46	12.52	12.57	8.81	8.25	7.91	6.41	5.48	5.97	6.15	7.26	9.11	11.20	
	HDD50	4593	965	861	697	270	65	2	0	1	51	288	579	814	
	HDD65	8797	1430	1281	1162	696	391	174	57	158	399	741	1029	1279	
	CDD50	1414	0	0	0	24	144	307	487	337	103	12	0	0	
	CDD65	143	0	0	0	0	5	29	79	29	1	0	0	0	
	CDH74	1198	0	0	0	1	60	237	627	266	7	0	0	0	
	CDH80	276	0	0	0	0	9	46	150	71	0	0	0	0	
Wind	WSAvg	4.4	4.7	4.7	5.0	4.5	4.8	4.0	3.4	3.5	3.9	4.8	5.0	5.0	
Precipitation	PrecAvg	28.0	2.0	1.5	1.5	1.3	2.4	3.2	3.6	3.3	2.5	2.6	2.1	2.0	
	PrecMax	33.8	3.2	3.1	2.7	2.4	5.2	5.6	7.4	7.1	5.0	5.6	4.1	3.6	
	PrecMin	21.7	0.4	0.4	0.5	0.3	0.7	0.6	0.7	0.4	0.6	0.5	0.0	0.7	
	PrecStd	3.1	0.7	0.7	0.5	0.6	1.2	1.4	1.9	1.7	1.1	1.3	0.9	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	40.0	52.6	71.9	79.7	85.0	89.0	88.2	74.2	64.8	49.9	45.5	
		MCWB	37.9	37.2	42.4	53.9	63.5	66.6	70.8	69.1	62.5	55.6	48.0	44.3	
	2%	DB	35.7	37.2	46.3	66.7	74.7	80.2	83.9	80.2	68.9	56.8	46.4	41.6	
		MCWB	34.8	35.1	39.4	50.9	59.4	65.7	68.0	65.9	59.5	51.9	44.8	40.5	
	5%	DB	33.9	35.3	42.1	61.1	70.3	76.2	80.5	76.0	65.1	53.9	43.8	38.3	
		MCWB	32.9	33.5	37.3	49.5	57.3	63.6	67.0	64.4	57.5	50.8	42.3	37.2	
	10%	DB	32.8	33.8	38.7	56.7	66.4	72.4	77.1	72.0	61.8	51.5	41.4	35.8	
		MCWB	31.9	32.6	35.3	47.1	55.1	61.4	65.3	62.3	55.9	48.9	40.1	34.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.2	38.3	44.5	56.7	65.2	69.6	72.5	71.2	64.3	56.4	48.3	44.4
			MCDB	38.7	39.5	50.6	69.9	78.0	80.1	83.4	85.5	71.6	61.8	49.2	45.3
2%		WB	35.0	35.4	40.3	52.6	61.9	67.3	70.0	68.1	61.2	53.3	44.7	40.4	
		MCDB	35.5	36.5	44.1	62.9	70.9	76.1	80.5	77.1	66.9	55.8	45.6	41.3	
5%		WB	33.2	33.9	37.8	50.3	59.2	65.4	68.3	65.7	58.6	51.3	42.3	37.4	
		MCDB	33.7	34.9	41.2	59.7	67.1	74.3	77.9	72.9	63.6	54.0	43.3	38.2	
10%		WB	32.0	32.7	35.7	48.0	56.9	63.1	66.6	63.6	56.7	49.2	40.1	34.9	
		MCDB	32.8	33.7	38.3	55.9	63.8	70.3	75.1	70.5	61.1	51.4	41.1	35.6	
Mean Daily Temperature Range	5% DB	MDBR	8.8	11.7	13.9	16.9	17.1	15.5	16.7	15.6	13.9	9.4	6.8	7.2	
		MCDBR	6.2	7.7	18.0	25.5	24.0	21.7	22.9	23.2	20.1	13.5	8.4	6.3	
		MCWBR	6.3	6.5	11.9	13.4	11.2	9.3	9.6	10.6	11.6	9.2	7.5	6.2	
	5% WB	MCDBR	6.3	6.3	14.2	22.3	21.3	18.9	20.5	19.1	16.8	11.8	7.6	6.3	
		MCWBR	6.4	5.8	10.1	12.7	10.7	9.1	9.7	10.0	11.0	8.7	7.3	6.1	
Clear-Sky Solar Irradiance	taub	0.280	0.314	0.335	0.384	0.377	0.366	0.393	0.398	0.359	0.340	0.314	0.273		
	taud	2.135	2.114	2.174	2.250	2.378	2.431	2.366	2.342	2.425	2.398	2.241	2.133		
	Ebn at Noon	182	226	256	262	273	278	266	255	250	221	169	149		
	Edh at Noon	17	28	34	37	35	33	35	34	27	21	16	13		
All-Sky Solar Radiation	RadAvg	141	390	864	1255	1627	1677	1640	1308	831	363	126	80		
	RadStd	30	64	93	138	150	169	195	132	102	75	22	23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page